snow.sno

Purpose

`snow.sno` is the snow parameter database. Each row defines a named set of snow accumulation and melt parameters. HRUs reference a row by name through the `snow` column of `hru-data.hru`, which the reader resolves to an index into the `snodb` array. If `hru-data.hru` carries `null` for an HRU's snow entry, that HRU uses the default `snow_parameters` values.

Source

- Reader: `src/snowdb_read.f90`
- Type definition: `src/hru_module.f90`, `type snow_parameters` (the `snodb` array)
- Filename slot: `in_parmdb%snow` in `src/input_file_module.f90` (default `snow.sno`).

Format

List-directed reads (free format, whitespace-separated). The first two lines are skipped. From line three to end of file each line is one snow parameter row. The reader counts records on a first pass, allocates `snodb(0:imax)`, then reads each row into `snodb(isno)`.

- Line 1: title (skipped). Editor convention: `snow.sno: written by SWAT+ editor ...`.
- Line 2: column header (skipped). Editor convention: `name fall_tmp melt_tmp melt_max melt_min tmp_lag snow_h2o cov50 snow_init`.
- Line 3+: one row per parameter set, 9 fields each.

The fields appear in the order declared in `type snow_parameters`:

#	Field	Type	Units	Default	Description
1	name	char(40)	-	(none)	Parameter set name. Referenced from the `snow` column of `hru-data.hru`.
2	falltmp	real	deg C	0.0	Snowfall temperature. Precipitation falls as snow when the mean air temperature is at or below this threshold.
3	melttmp	real	deg C	0.5	Snowmelt base temperature. Snowmelt only occurs when the snowpack temperature exceeds this value.
4	meltmx	real	mm/deg C/day	4.5	Maximum melt rate factor for snow during the year (occurs near 21 June).
5	meltmn	real	mm/deg C/day	0.5	Minimum melt rate factor for snow during the year (occurs near 21 December).
6	timp	real	none	0.8	Snowpack temperature lag factor (0 to 1). Higher values give more weight to the current day's air temperature.
7	covmx	real	mm H2O	25.0	Snow water content above which the ground is fully covered by snow.
8	cov50	real	none	0.5	Fraction of `covmx` at which the ground is 50% snow-covered. Shape factor for the snowpack areal depletion curve.
9	init_mm	real	mm H2O	0.0	Initial snow water content at the start of the simulation.

The defaults above are the values assigned in `type snow_parameters` at declaration. They are not values the reader supplies, so every row of `snow.sno` must list all nine fields.

Example

`refdata/Ames_sub1/snow.sno`:

```
snow.sno: written by SWAT+ editor v2.3.3 on 2023-10-25 08:58 for SWAT+ rev.60.5.7
name                   fall_tmp      melt_tmp      melt_max      melt_min       tmp_lag      snow_h2o
snow001                 1.00000       0.50000       4.50000       4.50000       1.00000       1.00000
```

One parameter set named `snow001`. Note that the example value for `meltmn` (column 5) is `4.5`, identical to `meltmx`. The defaults shipped in source set `meltmn = 0.5`.

Related files

- `hru-data.hru`. References a `snow.sno` row by name in its `snow` column. `null` selects the type defaults.
- `file.cio`. Names the `snow.sno` file in the `init` group via `in_parmdb%snow`. Set to `null` to skip reading (no snow database is loaded; HRUs that point at a snow name will fall back as described above).